

Computer Science 2210 – Quiz 3

October 26, 2011

Grade: 86

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1. Beside each part below, write the letter that indicates the correct **Big-Oh** representation of the number of operations (amount of work) required to solve the specified problem. Your possible choices are:

- A. $O(1)$ – essentially constant time B. $O(N)$ C. $O(N^2)$ D. $O(N^3)$
E. $O(\log_2(N))$ F. $O(N \log_2(N))$ G. $O(N!)$ H. None of these

- a. B Performing a sequential search of an unsorted array of integers with N entries
b. B Calculating $N!$
c. B Computing the average of N doubles
d. A Computing $N^3 + 4N^2 + 17N + 243$
e. B Evaluating 3^N
f. A Finding the largest value in a sorted array of N integers
g. B Determining the smallest value in an unsorted array of N decimal values
h. B Locating the alphabetically last string in an array of N unsorted strings
i. E Performing a binary search of a sorted array of N strings
j. C Performing a sequential search of a **2-D** array of integers with N rows and N columns
k. B Performing a sequential search of a **2-D** array of integers with N rows and 4 columns
l. A Sequentially searching an unsorted array of 1000 strings
m. D Processing a **3-level nested loop**, with the outer executed up to $N-1$ times, the middle executed up to N times, and the inner executed up to $N-1$ times
n. C Sorting an array of N random integers using a **Selection Sort** algorithm
o. C Sorting an array of N random doubles using the **Insertion Sort**
p. Execution of the following code:

```
for (int i = 0; i < 10; i++)
    for (int j = 0; j < N; j++)
        for (int k = 0; k < 25; k++)
            Console.WriteLine (i * j * k);
```

B

2. Suppose an upper bound on the number of operations required by a particular algorithm is expressed by the following formula in terms of the number of data items it processes: $f(N) = 7N^3 + 24N^2 - 103N + N \log_2(N) + 1000$. The proper **Big-Oh** notation describing this fact is $O(N^3)$.

3. Circle each of the following that we discussed in class as being a common feature of a **code-profiler**.

- ☒ a. Measures the processing time required to execute a given method.
☐ b. Counts the number of lines of source code in a function.
☒ c. Helps identify methods that a particular test case did not test.
☒ d. Helps to identify bottlenecks in execution of an algorithm.
☒ e. Helps identify parts of a program for which efficiency improvement will not pay significant dividends.
☐ f. Used to identify and correct syntax errors.
☒ g. Shows the number of times a given method is invoked.

4. When comparing algorithms, areas of interest may include run-time efficiency, memory and other resource usage efficiency, and ease of programming, testing, and maintenance of the code. Circle the item describing a common relationship between any two of these?

- a. Direct c. Reverse e. Converse
☒ b. Inverse d. Transverse f. Ternary

Points deducted: 4

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5. For each of the following answer T for True or F for False.

- a. F A **tag sort** exchanges the data itself instead of references to the data such as subscripts
- b. F **Big-Oh** notation describes the performance of an algorithm for any number of data items
- c. F An exponential algorithm is acceptable if $N < 1000$ (approximately).
- d. F An algorithm requiring **10N** units of work needs 5 orders of magnitude more operations than one requiring **2N** units of work
- e. F The median-of-three quicksort performs better than any other sorting algorithm in all cases
- f. F The Shell sort with a gap determined by dividing by 2.2 each time performs better than any other sorting algorithm in all cases
- g. F Any algorithm that is $O(N^2)$ is also $O(N^3)$
- h. F For large values of N , $N \log_2(N) > N^2$
- i. T For large values of N , $2^N > N^2$
- j. F On a large sorted array, a **binary search** is ALWAYS faster than a **sequential search**

6. For an array of **1000** integers containing the values **0 – 999** in that order, answer the following questions.

- a. The **minimum** number of comparisons required by a **sequential search** is 1. This number of comparisons is required if the **value** being sought is 0.
- b. The **maximum** number of comparisons required by a **sequential search** is 999.
- c. The **minimum** number of comparison required by a **binary search** is 1. This number of comparisons is required if the **value** being sought is 500.
- d. The **maximum** number of comparisons required by a **binary search** is 10.

7. Consider an array of integers with the following values in this order: **7 2 8 6 3 1 9 4 5**

- a. After the first exchange of values in a **Selection** sort, the values are: 1 7 2 8 6 3 9 4 5
- b. After the first pass of a **Sinking** sort, the values are: 2 7 6 3 1 8 4 5 9
- c. After one pass with a gap of 3 in a **Shell** sort, the values are: 6 2 1 7 3 5 9 4 8
- d. Using the **median-of-three** technique to determine a **pivot** for the **Quick Sort**, the “three” are 7, 3, and 9 and the median is 3.

8. **Big-Oh** is used to designate the order of magnitude of the work done by an algorithm on a large number of data items in (Circle One):

- | | | |
|--------------------------|---------------------|------------------|
| a. the best case | c. the average case | e. one case |
| <u>b.</u> the worst case | d. every case | f. none of these |

9. To perform a **binary search** on a **List** of data type **T** that implements **IEquatable<T>**, what must be true of the data in the **List**? must also be comparable

10. The sorting algorithm we discussed being similar to how we arrange cards as they are dealt is insertion sort

11. The sorting algorithm that finds the largest value and exchanges it directly with the last item is selection sort